Module 1: Sequences, Series, and Power Series

1. Sequences, Examples
2. Convergence and Divergence of Sequences, Examples
3. Series, Examples, Convergence of Series
4. Tests of Convergence of Series
5. Power Series and its convergence

Sequences

1. Sequence: a list of objects.
2. We will only consider (real) number sequences.
3. Our sequences will be **infinite**, like $1, 3, 5, \ldots, (2n-1), \ldots$.

Applications:

1. Study of dynamical systems.
2. Stability analysis of physical processes.
3. Convergence of (infinite) series.

Examples

- $1, 2, 3, \ldots, n, \ldots$
- $2, 3, 5, 7, 11, \ldots, p_n, \ldots$
- $1, 1, 1, \ldots, 1, \ldots$
- $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{n}, \ldots$
- $0, 1, 1, 2, 3, 5, 8, 13, \ldots, \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right], \ldots$
- $\sqrt{1}, \sqrt{2}, \sqrt{3}, \sqrt{4}, \ldots, \sqrt{n}, \ldots$

Sequence: Formal Definition

A sequence is an infinite list of numbers:

$$a_1, a_2, a_3, \ldots, a_n, \ldots$$

We can also view it as a function!

Definition

A *sequence* is a function from the set of positive integers to the set of real numbers:

$$a : \mathbb{Z}^+ \to \mathbb{R}.$$

The images $a(1), a(2), a(3), \ldots, a(n), \ldots$ are called the *terms* of the sequence.

Note

1. The images $a(1), a(2), a(3), \ldots, a(n), \ldots$ are often denoted by $a_1, a_2, a_3, \ldots, a_n, \ldots$.
2. The sequence $a_1, a_2, a_3, \ldots, a_n, \ldots$ is often denoted by $\{a_n\}$: i.e., the nth term of the sequence is enclosed within braces.

So,

$$\{a_n\} = a_1, a_2, a_3, \ldots, a_n, \ldots.$$

Examples

$$
\begin{aligned}
\{n\} &= 1, 2, 3, \ldots, n, \ldots \\
\left\{\frac{1}{n}\right\} &= 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{n}, \ldots \\
\left\{\frac{n-1}{n}\right\} &= 0, \frac{1}{2}, \frac{2}{3}, \ldots, \frac{n-1}{n}, \ldots \\
\{1\} &= 1, 1, 1, \ldots, 1, \ldots \\
\left\{(-1)^{n+1}\right\} &= 1, -1, 1, -1, 1, \ldots, (-1)^{n+1}, \ldots \\
\left\{\frac{(-1)^{n+1}}{n}\right\} &= 1, -\frac{1}{2}, \frac{1}{3}, \ldots, \frac{(-1)^{n+1}}{n}, \ldots \\
\left\{(-1)^{n+1} n\right\} &= 1, -2, 3, -4, \ldots, (-1)^{n+1} n, \ldots \\
\{\sqrt{n}\} &= \sqrt{1}, \sqrt{2}, \sqrt{3}, \sqrt{4}, \ldots, \sqrt{n}, \ldots
\end{aligned}
$$

Note: Different sequences behave differently.

Convergence of a Sequence

Definition (Converges, Diverges, Limit)

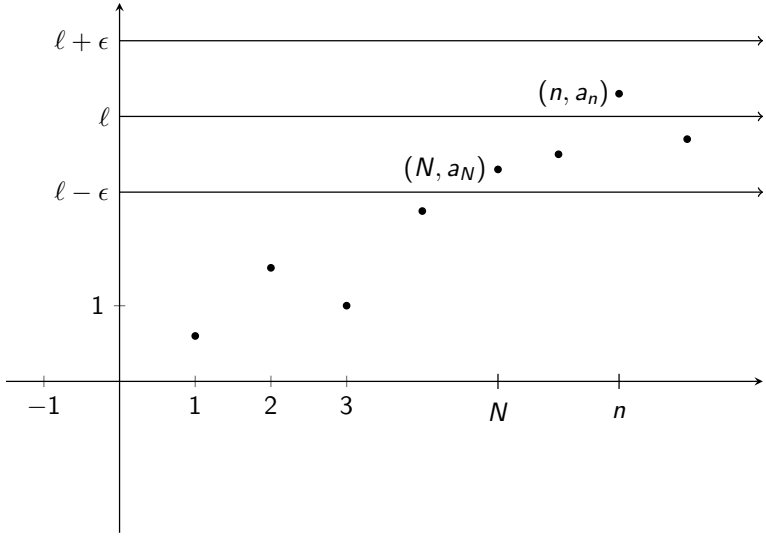
A sequence $\{a_n\}$ *converges* to a number ℓ if to every positive number ϵ, there corresponds an integer N such that for all $n \geq N$,

$$|a_n - \ell| < \epsilon.$$

If no such number ℓ exists, we say that $\{a_n\}$ *diverges*.

If $\{a_n\}$ converges to ℓ, we write $\lim_{n\to\infty} a_n = \ell$ or $a_n \to \ell$ and call ℓ the *limit* of the sequence.

Convergence, Pictorially



Applying the Definition

Show using the definition of convergence that

$$\lim_{n\to\infty} \frac{1}{n} = 0.$$

Solution: Here $a_n = \frac{1}{n}$ and $\ell = 0$.
Let $\epsilon > 0$ be given (fixed). We must find a fixed integer N such that, for $n \geq N$,

$$\left|\frac{1}{n} - 0\right| < \epsilon.$$

Now,

$$\left|\frac{1}{n} - 0\right| < \epsilon \iff \frac{1}{n} < \epsilon \iff n > \frac{1}{\epsilon}.$$

Let N be any integer greater than $\frac{1}{\epsilon}$.

Let N be any integer greater than $\frac{1}{\epsilon}$.

Then $n \geq N > \frac{1}{\epsilon} \quad \Rightarrow \quad \left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \epsilon.$

Classwork

Prove that the constant sequence

$$\{3\} = 3, 3, 3, \ldots, 3, \ldots$$

converges to 3.

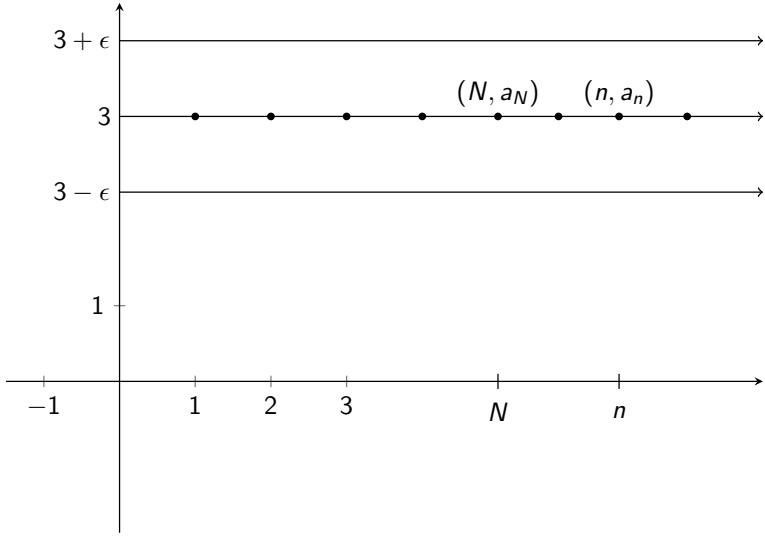
Solution: Here $a_n = 3$ and $\ell = 3$.

Let $\epsilon > 0$ be given. We must find a fixed integer N such that, for all $n \in \mathbb{N}$,

$$n \geq N \implies |3 - 3| < \epsilon.$$

Choose any integer $N \geq 1$. It works!

Example, Pictorially



Homework

For any integer $n \geq 2$,

$$1 < n^{\frac{1}{n}} < 2.$$

Example

Prove that $n^{\frac{1}{n}} \longrightarrow 1$.

Solution: Here $a_n = n^{\frac{1}{n}}$ and $\ell = 1$.

Let $\epsilon > 0$ be given. We must find a fixed integer N such that, for $n \in \mathbb{N}$,

$$n \geq N \quad \Rightarrow \quad \left| n^{\frac{1}{n}} - 1 \right| < \epsilon.$$

Let

$$h_n = \left| n^{\frac{1}{n}} - 1 \right|.$$

If $n = 1$, then $h_n = 0$.

We have $h_n = \left| n^{\frac{1}{n}} - 1 \right|$. If $n = 1$, then $h_n = 0$.

Suppose $n \geq 2$. Then

$$h_n = n^{\frac{1}{n}} - 1 \quad \text{and} \quad h_n > 0$$

since $1 < n^{\frac{1}{n}} < 2$ (homework).

So,

$$n^{\frac{1}{n}} = 1 + h_n \quad \text{or} \quad n = (1 + h_n)^n.$$

$$
\begin{aligned}
n &= (1+h_n)^n \\
&= 1 + \binom{n}{1} h_n + \binom{n}{2} h_n^2 + \ldots + h^n \\
&> \binom{n}{2} h_n^2 \\
&= \frac{n(n-1)}{2} h_n^2
\end{aligned}
$$

Thus

$$
\frac{n-1}{2} h_n^2 < 1 \quad \text{or} \quad h_n < \sqrt{\frac{2}{n-1}}.
$$

$$h_n < \sqrt{\frac{2}{n-1}}.$$

That is,

$$\left| n^{\frac{1}{n}} - 1 \right| < \sqrt{\frac{2}{n-1}}.$$

Our goal: To find an integer N such that, for all $n \in \mathbb{N}$,

$$n \geq N \Longrightarrow \left| n^{\frac{1}{n}} - 1 \right| < \epsilon.$$

But we now find an integer N such that, for all $n \in \mathbb{N}$,

$$n \geq N \Longrightarrow \sqrt{\frac{2}{n-1}} < \epsilon.$$

Now,

$$\begin{aligned}
\sqrt{\frac{2}{n-1}} < \epsilon \quad &\Leftrightarrow \quad \frac{2}{n-1} < \epsilon^2 \\
&\Leftrightarrow \quad \frac{n-1}{2} < \frac{1}{\epsilon^2} \\
&\Leftrightarrow \quad n > 1 + \frac{2}{\epsilon^2}
\end{aligned}$$

Choose any integer $N > 1 + \frac{2}{\epsilon^2}$.

Choose any integer $N > 1 + \frac{2}{\epsilon^2}$.

$$\begin{aligned} n \geq N \quad &\Rightarrow \quad n > 1 + \frac{2}{\epsilon^2} \\ &\Rightarrow \quad \frac{2}{n-1} < \epsilon \\ &\Rightarrow \quad \sqrt{\frac{2}{n-1}} < \epsilon \\ &\Rightarrow \quad h_n < \epsilon \\ &\Rightarrow \quad \left| n^{\frac{1}{n}} - 1 \right| < \epsilon \end{aligned}$$

Hence

$$n^{\frac{1}{n}} \longrightarrow 1.$$

Theorem

Let $\{a_n\}$ be a convergent sequence. Then its limit is unique.

Proof:

- Suppose $\{a_n\}$ converges to both ℓ_1 and ℓ_2.
- Let $\epsilon > 0$ be any positive real number.
- Then, corresponding to $\epsilon/2 > 0$, we can find integers N_1 and N_2 such that

$$n \geq N_1 \Longrightarrow |a_n - \ell_1| < \frac{\epsilon}{2};$$

$$n \geq N_2 \Longrightarrow |a_n - \ell_2| < \frac{\epsilon}{2}.$$

- Let $N = \max(N_1, N_2)$.
- Then, for $n \geq N$, we have

$$|\ell_1 - \ell_2| = |(a_n - \ell_1) - (a_n - \ell_2)| \leq |a_n - \ell_1| + |a_n - \ell_2| < \epsilon.$$

We have

$$|\ell_1 - \ell_2| < \epsilon.$$

But ϵ is arbitrary.

So, we have that $\ell_1 = \ell_2$.

Divergent Sequences: Example

Show that the sequence $1, -1, 1, -1, \ldots, (-1)^{n+1}, \ldots$ diverges.
Solution:

- Suppose the sequence converges to some number ℓ.
- Let $\epsilon = 1/2$.
- Then there must be an integer N such that $|a_n - \ell| < 1/2$ for all $n \geq N$. That is $\ell - 1/2 < a_n < \ell + 1/2$.
- As the number 1 appears repeatedly as every other term, we must have $\ell - 1/2 < 1 < \ell + 1/2$. This implies that $1/2 < \ell < 3/2$.
- Similarly, as the number -1 appears repeatedly as every other term, we must have $\ell - 1/2 < -1 < \ell + 1/2$. This implies that $-1/2 < \ell < -3/2$.
- Thus we have that the number ℓ lies in both of the intervals $(1/2, 3/2)$ and $(-3/2, -1/2)$. But this is impossible. Hence, our assumption is wrong and the sequence diverges.

Divergent Sequences

Sequences that are diverging to infinity:

$$\{n\},\ \{n^2\},\ \{\sqrt{n}\},\ \{\log n\} \text{ etc.}$$

Divergence to Infinity

Definition (Diverges to Infinity)

A sequence $\{a_n\}$ **diverges to infinity** if for every number M, there is an integer N such that for all n with $n \geq N$, $a_n > M$. In this case, we write

$$\lim_{n\to\infty} a_n = \infty \quad \text{or} \quad a_n \to \infty.$$

Homework: Provide a definition for the divergence of a sequence to $-\infty$.

Theorem

Let $\{a_n\}$ *and* $\{b_n\}$ *be convergent sequences with* $\lim_{n\to\infty} a_n = a$ *and* $\lim_{n\to\infty} b_n = b$. *Then*

1. $\lim_{n\to\infty} (a_n + b_n) = a + b.$
2. $\lim_{n\to\infty} (a_n - b_n) = a - b.$
3. $\lim_{n\to\infty} (ka_n) = ka.$ *(Any number k).*
4. $\lim_{n\to\infty} (a_n b_n) = ab.$
5. $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{a}{b}$ *if* $b \neq 0.$

Proof of (1):

- Let $\epsilon > 0$ be given.
- Since $\lim_{n\to\infty} a_n = a$ and $\lim_{n\to\infty} b_n = b$, corresponding to $\epsilon/2 > 0$, there are integers N_1 and N_2 such that

$$n \geq N_1 \Rightarrow |a_n - a| < \frac{\epsilon}{2};$$

$$n \geq N_2 \Rightarrow |b_n - b| < \frac{\epsilon}{2}.$$

- Let $N = \max(N_1, N_2)$.
- Then for $n \geq N$, we have

$$\begin{aligned} |(a_n + b_n) - (a + b)| &= |(a_n - a) + (b_n - b)| \\ &\leq |a_n - a| + |b_n - b| < \tfrac{\epsilon}{2} + \tfrac{\epsilon}{2} = \epsilon. \end{aligned}$$

- This proves Part 1, that is, $\lim_{n\to\infty} (a_n + b_n) = a + b$.

Proof of (4):

Part 2 of the theorem implies the following:

Fact

$$\lim_{n\to\infty} a_n = a \Leftrightarrow \lim_{n\to\infty} (a_n - a) = 0.$$

Consider the following identity:

$$a_n b_n - ab = (a_n - a)(b_n - b) + a(b_n - b) + b(a_n - a).$$

From the fact above and Part 3 of the theorem, we have that

$$\lim_{n\to\infty} [a(b_n - b)] = 0 \quad \text{and} \quad \lim_{n\to\infty} [b(a_n - a)] = 0.$$

We will now prove that $\lim_{n\to\infty}[(a_n - a)(b_n - b)] = 0$.

Proof of (4):

Given $\epsilon > 0$, corresponding to $\sqrt{\epsilon} > 0$, there are integers N_1 and N_2 such that

$$n \geq N_1 \Rightarrow |a_n - a| < \sqrt{\epsilon};$$

$$n \geq N_2 \Rightarrow |b_n - b| < \sqrt{\epsilon}.$$

Let $N = \max(N_1, N_2)$. Then

$$n \geq N \Rightarrow |(a_n - a)(b_n - b)| < \epsilon.$$

This proves that $\lim_{n\to\infty}[(a_n - a)(b_n - b)] = 0$.
Thus the fact and the identity in the preceding slide imply that

$$\lim_{n\to\infty} (a_n b_n) = ab.$$

Examples

(a) $\lim_{n\to\infty}\left(-\frac{1}{n}\right) = -1\cdot\lim_{n\to\infty}\frac{1}{n} = -1\cdot 0 = 0.$

(b) $\lim_{n\to\infty}\frac{n-1}{n} = \lim_{n\to\infty}\left(1-\frac{1}{n}\right) = \lim_{n\to\infty}1 - \lim_{n\to\infty}\frac{1}{n} = 1-0 = 1.$

(c) $\lim_{n\to\infty}\frac{5}{n^2} = 5\cdot\lim_{n\to\infty}\frac{1}{n}\cdot\lim_{n\to\infty}\frac{1}{n} = 5\cdot 0\cdot 0 = 0.$

Homework

1. Prove the other parts of the preceding theorem.
2. Prove that $\lim_{n\to\infty} \dfrac{4-7n^6}{n^6+3} = -7$.
3. Prove that if the sequence $\{a_n\}$ diverges and c is any nonzero constant, then the sequence $\{ca_n\}$ also diverges.

Theorem (The Sandwich Theorem for Sequences)

Let $\{a_n\}, \{b_n\}, \{c_n\}$ be sequences of real numbers. If $a_n \leq b_n \leq c_n$ for all n beyond some index N and if $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = \ell$, then $\lim_{n\to\infty} b_n = \ell$.

Proof.

Homework! □

Corollary

If $|b_n| \leq c_n$ *and* $c_n \to 0$*, then* $b_n \to 0$.

Proof.

$$|b_n| \leq c_n \Leftrightarrow -c_n \leq b_n \leq c_n.$$

$c_n \to 0$. So, by the Sandwich theorem, we have $b_n \to 0$. □

Applying the Sandwich Theorem: Examples

$\frac{1}{n} \to 0$. So, we have the following true:

(a) $\frac{\cos n}{n} \to 0$ since $-\frac{1}{n} \leq \frac{\cos n}{n} \leq \frac{1}{n}$.

(b) $\frac{1}{2^n} \to 0$ since $0 \leq \frac{1}{2^n} \leq \frac{1}{n}$.

(c) $(-1)^n \frac{1}{n} \to 0$ since _____ ?

Theorem (The Continuous Function Theorem for Sequences)

Let $\{a_n\}$ be a sequence of real numbers. If $a_n \to \ell$ and if f is a function that is continuous at ℓ and is defined at all a_n, then $f(a_n) \to f(\ell)$.

Applying the Continuous Function Theorem: Examples

1. Show that $\sqrt{(n+1)/n} \to 1$.

   Solution: We know that $\dfrac{n+1}{n} \to 1$. Take $a_n = (n+1)/n$, $f(x) = \sqrt{x}$ and $\ell = 1$. The function $f(x)$ is continuous at $\ell = 1$. Thus by the theorem, $\sqrt{(n+1)/n} \to \sqrt{1} = 1$.

2. Show that $2^{1/n} \to 1$.

   Solution: The sequence $\left\{\dfrac{1}{n}\right\}$ converges to 0. Take $a_n = 1/n$ and $f(x) = 2^x$ and $\ell = 0$.
   The function is continuous at $\ell = 0$.
   Then we have that $2^{1/n} \to 2^0 = 1$.

Theorem

Suppose that $f(x)$ is a function defined for all $x \geq n_0$ and that $\{a_n\}$ is a sequence of real numbers such that $a_n = f(n)$ for all $n \geq n_0$. If

$$\lim_{x \to \infty} f(x) = \ell,$$

then

$$\lim_{n \to \infty} a_n = \ell.$$

Applying l'Hôpital's Rule: Examples

Show that

$$\lim_{n\to\infty} \frac{\ln n}{n} = 0.$$

Solution: The function $f(x) = \dfrac{\ln x}{x}$ is defined for all $x \geq 1$ and agrees with the given sequence at positive integers.
Thus by the preceding theorem $\lim_{n\to\infty} \frac{\ln n}{n}$ will equal $\lim_{x\to\infty} \frac{\ln x}{x}$ if the latter limit exists.

Applying the l'Hôpital's rule, we see that

$$\lim_{x\to\infty} \frac{\ln x}{x} = \lim_{x\to\infty} \frac{1/x}{1} = \frac{0}{1} = 0.$$

Thus we conclude that $\lim\limits_{n\to\infty} \dfrac{\ln n}{n} = 0.$

Applying l'Hôpital's Rule: Examples

Find

$$\lim_{n\to\infty} \frac{5^n}{7n}.$$

Solution: By l'Hôpital's rule,

$$\lim_{n\to\infty} \frac{5^n}{7n} = \lim_{x\to\infty} \frac{5^x}{7x} = \lim_{x\to\infty} \frac{5^x \ln 5}{7} = \infty.$$

Applying l'Hôpital's Rule: Homework

Does the sequence whose nth term is

$$a_n = \left(\frac{n+1}{n-1}\right)^n$$

converge? If so, find $\lim_{n\to\infty} a_n$.

Note: Here the limit leads to the indeterminate form 1^∞. We can apply l'Hôpital's rule if we first change the form to $\infty \cdot 0$ by taking the natural logarithm of a_n.

Some Special Sequences

1. If $p > 0$, then $\lim_{n\to\infty} \frac{1}{n^p} = 0$.
2. If $p > 0$, then $\lim_{n\to\infty} \sqrt[n]{p} = 1$.
3. $\lim_{n\to\infty} \sqrt[n]{n} = 1$.
4. If $p > 0$ and α is real, then $\lim_{n\to\infty} \frac{n^\alpha}{(1+p)^n} = 0$.
5. If $|x| < 1$, then $\lim_{n\to\infty} x^n = 0$.
6. $\lim_{n\to\infty} \frac{x^n}{n!} = 0$.
7. $\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$.

Proof of (1)

We want to prove: If $p > 0$, then $\lim_{n\to\infty} \frac{1}{n^p} = 0$.

Let $\epsilon > 0$ be a fixed number. We want to find an integer N such that for all $n \geq N$,

$$\left| \frac{1}{n^p} - 0 \right| < \epsilon.$$

$$\Leftrightarrow \frac{1}{n^p} < \epsilon$$

$$\Leftrightarrow \frac{1}{n} < \epsilon^{1/p}$$

$$\Leftrightarrow n > (1/\epsilon)^{1/p}.$$

Proof of (1) continued

Choose N to be any positive integer bigger than $(1/\epsilon)^{1/p}$. Then for $n \geq N$, we have

$$n \geq N > (1/\epsilon)^{1/p}$$

$$\Leftrightarrow \frac{1}{n} < \epsilon^{1/p}$$

$$\Leftrightarrow \frac{1}{n^p} < \epsilon$$

$$\Leftrightarrow \left| \frac{1}{n^p} - 0 \right| < \epsilon.$$

This proves that the sequence $\left\{\frac{1}{n^p}\right\}$ converges to 0.

Proof of (2)

We want to prove: If $p > 0$, then $\lim_{n\to\infty} \sqrt[n]{p} = 1$.

We divide our proof into three cases.

First assume that $p > 1$. Let $h_n = \sqrt[n]{p} - 1$. Then $h_n > 0$. Now, by the Binomial theorem, we have

$$p = (1 + h_n)^n \geq 1 + nh_n$$

$$\Leftrightarrow h_n \leq \frac{p-1}{n}.$$

But, we also have $h_n > 0$. Thus,

$$0 < h_n \leq \frac{p-1}{n}.$$

As $\lim_{n\to\infty} 0 = 0$ and $\lim_{n\to\infty} \frac{p-1}{n} = 0$, by the Sandwich theorem, $\lim_{n\to\infty} h_n = 0$. This implies that

$$\lim_{n\to\infty} \sqrt[n]{p} = 1.$$

Proof of (2) continued

Now, assume that $p = 1$. Then the sequence $\{a_n\}$ is the constant sequence $\{1\}$, whose limit is 1.

Lastly, assume that $0 < p < 1$. Let $q = \frac{1}{p}$. Then $q > 1$. By the above proof, we have

$$\lim_{n\to\infty} q^{1/n} = 1$$

$$\Rightarrow \lim_{n\to\infty} \left(\frac{1}{p}\right)^{1/n} = 1$$

$$\Rightarrow \frac{1}{\lim_{n\to\infty} p^{1/n}} = 1$$

$$\Rightarrow \lim_{n\to\infty} p^{1/n} = 1.$$

Proof of (3)

We want to prove: $\lim_{n\to\infty} \sqrt[n]{n} = 1$.

Let $h_n = \sqrt[n]{n} - 1$. For $n \geq 2$, by the binomial theorem, we have

$$n = (1 + h_n)^n \geq \frac{n(n-1)}{2} h_n^2$$

$$\Leftrightarrow h_n \leq \sqrt{\frac{2}{n-1}}.$$

We also have $h_n > 0$ for $n \geq 2$. Therefore,

$$0 < h_n \leq \sqrt{\frac{2}{n-1}} \quad \text{for } n \geq 2.$$

Proof of (3) continued

As $\lim_{n\to\infty} 0 = 0$ and $\lim_{n\to\infty} \sqrt{\frac{2}{n-1}} = 0$, by the Sandwich theorem,

$$\lim_{n\to\infty} h_n = 0.$$

This implies that

$$\lim_{n\to\infty} \sqrt[n]{n} = 1.$$

Proof of (4)

We want to prove: If $p > 0$ and α is a real number, then $\lim_{n\to\infty} \dfrac{n^\alpha}{(1+p)^n} = 0$.

Let k be an integer such that $k > \alpha$ and $k > 0$. For $n > 2k$, by the binomial theorem,

$$(1+p)^n > \binom{n}{k} p^k = \frac{n(n-1)\cdots(n-k+1)}{k!} p^k > \frac{n^k p^k}{2^k k!}.$$

Therefore,

$$0 < \frac{n^\alpha}{(1+p)^n} < \frac{2^k k!}{p^k} \frac{1}{n^{k-\alpha}} \qquad (\text{for } n > 2k).$$

Proof of (4) continued

As $k - \alpha > 0$, by part (1), $\frac{1}{n^{k-\alpha}} \to 0$. Therefore,

$$\frac{2^k k!}{p^k} \frac{1}{n^{k-\alpha}} \to 0.$$

Hence, by the sandwich theorem

$$\frac{n^\alpha}{(1+p)^n} \to 0.$$

Bounded Sequences

Definition

A sequence $\{a_n\}$ is said to be **bounded above** if there exists a number M such that $a_n \leq M$ for all n. In this case, M is called an **upper bound** for the sequence $\{a_n\}$.

If M is an upper bound for $\{a_n\}$ and no number less than M is an upper bound for $\{a_n\}$, then M is called the **least upper bound** for $\{a_n\}$.

Examples:

1. The sequence $1, 3, 5, \ldots$ has no upper bound.
2. The sequence $1, -1, 1, -1, \ldots$ is bounded above by $M = 1$. In fact, $M = 1$ is the least upper bound for this sequence.
3. The sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots, \frac{n}{n+1}, \ldots$ is bounded above by $M = 1$. In fact, $M = 1$ is the least upper bound for this sequence.

Bounded Sequences cont...

Definition

A sequence $\{a_n\}$ is said to be **bounded below** if there exists a number m such that $a_n \geq m$ for all n.

In this case, m is called a **lower bound** for the sequence $\{a_n\}$.

If m is a lower bound for $\{a_n\}$ and no number bigger than m is a lower bound for $\{a_n\}$, then m is called the **greatest lower bound** for $\{a_n\}$.

Definition

A sequence $\{a_n\}$ is said to be **bounded** if it is both bounded above and bounded below.

Homework

- Prove that a sequence $\{a_n\}$ is bounded if and only if, for some number K,

$$|a_n| \leq K \quad \text{for all } n.$$

- Prove that, if a sequence is convergent, then it is bounded.
- Prove that, the converse of the above (2nd result) is not true. That is, a bounded sequence need not converge.

Monotonic Sequences

Definition

A sequence $\{a_n\}$ is said to be

- **monotonically increasing** or **non-decreasing** if

$$a_n \leq a_{n+1} \quad \text{for all } n;$$

- **monotonically decreasing** or **non-increasing** if

$$a_n \geq a_{n+1} \quad \text{for all } n.$$

Examples of monotonic sequences

1. The sequence $1, 2, 3, \ldots, n, \ldots$ of positive integers is monotonically increasing.
2. The sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots, \frac{n}{n+1}, \ldots$ is monotonically increasing.
3. The sequence $2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \ldots, \frac{n+1}{n}, \ldots$ is monotonically decreasing.
4. The constant sequence $3, 3, 3, \ldots$ is both monotonically increasing and monotonically decreasing.

The Completeness Property of Real Numbers

A monotonically increasing sequence that is bounded above has a least upper bound.

And a monotonically decreasing sequence that is bounded below has a greatest lower bound.

Examples:

1. The sequence $0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots, \frac{n-1}{n}, \ldots$ is monotonically increasing and bounded above. It has 1 as its least upper bound.
2. The sequence $1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}, \ldots$ is monotonically decreasing and bounded below. It has 0 as its greatest lower bound.

Theorem

Let $\{a_n\}$ be a monotonically increasing sequence that is bounded above. Then it converges to its least upper bound.

Theorem

Let $\{a_n\}$ be a monotonically decreasing sequence that is bounded below. Then it converges to its greatest lower bound.

Proof:

- Let $\{a_n\}$ be a monotonically increasing sequence that is bounded above.
- Let ℓ be the least upper bound of the sequence. (This follows from the Completeness property of real numbers).
- Then we have that $a_n \leq \ell$ for all n.
- Let $\epsilon > 0$ be any real number.
- Then $\ell - \epsilon$ cannot be an upper bound for the sequence as $\ell - \epsilon < \ell$.
- Thus there is an integer N such that $a_N > \ell - \epsilon$.
- But $\{a_n\}$ is monotonically increasing. So, we have that

$$\text{if } n \geq N \text{ then } a_n \geq a_N > \ell - \epsilon.$$

- This implies that, for all $n \geq N$, $\ell - \epsilon < a_n < \ell + \epsilon$ equivalently, $|a_n - \ell| < \epsilon$.
- Hence a_n converges to ℓ.

Homework

- Prove that the sequence $\left\{\left(1+\frac{1}{n}\right)^n\right\}$ converges by showing that it is both monotonically increasing and bounded above.
- Let s be any positive number. Prove that the following recursively defined sequence $\{a_n\}$ is monotonically decreasing and bounded below. Also prove that it converges to $\sqrt{s}$.

$$a_1 = s \text{ and } a_{n+1} = \frac{1}{2}\left(a_n + \frac{s}{a_n}\right) \text{ for } n \geq 1.$$

Subsequences

Consider the sequence of positive integers

$$1, 2, 3, \ldots, n, \ldots$$

Some subsequences:

- $1, 2, 3, \ldots, n, \ldots$
- $1, 3, 5, \ldots, 2n - 1, \ldots$
- $2, 4, 6, \ldots, 2n, \ldots$
- $2, 3, 5, 7, \ldots, p_n, \ldots$

Subsequences

Definition

Let

$$\{a_n\} = a_1, a_2, a_3, \ldots, a_n, \ldots$$

be any sequence. If

$$\{n_k\} = n_1, n_2, \ldots, n_k, \ldots$$

is a sequence of positive integers such that

$$n_1 < n_2 < n_3, \ldots,$$

then

$$\{a_{n_k}\} = a_{n_1}, a_{n_2}, \ldots, a_{n_k}, \ldots$$

is called a subsequence of $\{a_n\}$.

Examples of subsequences

Consider the sequence of positive integers

$$\left\{\frac{1}{n}\right\} = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{n}, \ldots$$

Some subsequences:

- $\left\{\frac{1}{n}\right\} = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \ldots, \frac{1}{n}, \ldots$
- $\left\{\frac{1}{2n-1}\right\} = 1, \frac{1}{3}, \frac{1}{5}, \frac{1}{7}, \ldots, \frac{1}{2n-1}, \ldots$
- $\left\{\frac{1}{2n}\right\} = \frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \ldots, \frac{1}{2n}, \ldots$
- $\left\{\frac{1}{p_n}\right\} = \frac{1}{2}, \frac{1}{3}, \frac{1}{5}, \ldots, \frac{1}{p_n}, \ldots$

Important results concerning subsequences

Theorem

Let $\{a_n\}$ be any sequence.

1. *If $\{a_n\}$ converges to ℓ, then every subsequence of $\{a_n\}$ also converges to ℓ.*
2. *If the subsequences $\{a_{2n-1}\}$ and $\{a_{2n}\}$ both converges to ℓ, then the sequence $\{a_n\}$ also converges to ℓ.*

Proof.

Homework! □

Theorem (Bolzano–Weierstrass theorem)

Every bounded sequence $\{a_n\}$ has a convergent subsequence.

Cauchy Sequences

Definition

A sequence $\{a_n\}$ is said to be a Cauchy sequence if for every $\epsilon > 0$ there is an integer N such that for all $n \geq N, m \geq N$,

$$|a_n - a_m| < \epsilon.$$

Examples:

- The sequence $\left\{\frac{1}{n}\right\}$ is a Cauchy sequence.
- The sequence $\left\{\frac{n+1}{n}\right\}$ is a Cauchy sequence.
- The sequence $\left\{(-1)^{n+1}\right\}$ is not a Cauchy sequence.
- The sequence $\left\{n + \frac{(-1)^n}{n}\right\}$ is not a Cauchy sequence.

Cauchy Sequences continued

Theorem

A sequence converges if and only it is a Cauchy sequence.

Series

Consider a tiny frog which is initially at the point 0 of the number line. It makes successive rightward jumps along the number line as follows.

It jumps 1 unit in step 1.

It jumps $\frac{1}{2}$ units in step 2.

It jumps $\frac{1}{4}$ units in step 3.

It jumps $\frac{1}{8}$ units in step 4.

It jumps $\frac{1}{16}$ units in step 5.

$\vdots$

Thus the total distance traveled by the frog is

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

The total distance traveled by the frog is

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots$$

Partial sum		**Partial sum**	**Value**
First:	$s_1 = 1$	$2 - 1$	1
Second:	$s_2 = 1 + \frac{1}{2}$	$2 - \frac{1}{2}$	$\frac{3}{2}$
Third:	$s_3 = 1 + \frac{1}{2} + \frac{1}{4}$	$2 - \frac{1}{4}$	$\frac{7}{4}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
nth:	$s_n = 1 + \frac{1}{2} + \frac{1}{4} + \ldots + \frac{1}{2^{n-1}}$	$2 - \frac{1}{2^{n-1}}$	$\frac{2^n - 1}{2^{n-1}}$

The sequence $\{s_n\}$ of partial sums converges to 2.

In this case, we say that the series **converges** to 2.

And say that the **sum** of the series is 2.

Definition (Series, nth Term, Partial Sum, Converges, Sum)

Given a sequence $\{a_n\}$, an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

is called a **series**. The number a_n is called the ***n*th term** of the series. The sequence $\{s_n\}$ defined by

$$\begin{aligned} s_1 &= a_1 \\ s_2 &= a_1 + a_2 \\ s_3 &= a_1 + a_2 + a_3 \\ &\vdots \\ s_n &= a_1 + a_2 + a_3 + \cdots + a_n \\ &\vdots \end{aligned}$$

is called the **sequence of partial sums** of the series and the number s_n is called the ***n*th partial sum**.

Definition *Contd.*

If the sequence $\{s_n\}$ of partial sums converges to a limit ℓ, we say that the series **converges** and that its **sum** is ℓ. In this case, we also write

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots = \sum_{n=1}^{\infty} a_n = \ell.$$

If the sequence of partial sums of the series does not converge, we say that the series **diverges**.

Example: Geometric Series

A **geometric series** is a series of the form

$$a + ar + ar^2 + \cdots + ar^{n-1} + \cdots = \sum_{n=1}^{\infty} ar^{n-1}.$$

Here a and r are fixed real numbers and $a \neq 0$. The **common ratio** r can be positive or negative or even zero.

We will now prove that the geometric series converges for each r with $|r| < 1$ and it diverges otherwise.

Geometric Series

Case 1: $r = 1$**:** The nth partial sum of the geometric series is

$$s_n = a + a(1) + a(1^2) + \cdots + a(1^{n-1}) = na.$$

So, the sequence of partial sums diverges to ∞ or $-\infty$ depending on whether $a > 0$ or $a < 0$. Hence, in this case, the series diverges.

Case 2: $r = -1$**:** The nth partial sum of the geometric series is

$$s_1 = a,\ s_2 = a + a(-1) = 0,\ s_3 = a + a(-1) + a = a,\ s_4 = 0,\ \ldots$$

So, the sequence of partial sums diverges as it oscillates between a and 0. Hence, in this case too, the series diverges.

Geometric Series

Case 3: $|r| \neq 1$**:**

$$\begin{aligned} s_n &= a + ar + ar^2 + \cdots + ar^{n-1} \\ rs_n &= ar + ar^2 + \cdots + ar^{n-1} + ar^n \\ s_n - rs_n &= a - ar^n \\ s_n(1-r) &= a(1-r^n) \end{aligned}$$

$$s_n = \frac{a(1-r^n)}{1-r} \quad (r \neq 1)$$

Thus if $|r| < 1$, then $r^n \to 0$ as $n \to \infty$ and so $s_n \to \dfrac{a}{1-r}$;
if $|r| > 1$, then $|r^n| \to \infty$ and the series diverges.

Geometric Series: Summary

If $|r| < 1$, the geometric series $a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$ converges to $a/(1-r)$:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \qquad |r| < 1.$$

If $|r| \geq 1$, the series diverges.

Example

The geometric series with $a = 1/9$ and $r = 1/3$ is

$$\frac{1}{9} + \frac{1}{27} + \frac{1}{81} + \cdots = \frac{1/9}{1 - (1/3)} = \frac{1}{6}.$$

Homework: Find the sum of the geometric series

$$\sum_{n=0}^{\infty} \frac{(-1)^n 2}{5(3^n)} = \frac{2}{5} - \frac{2}{5 \cdot 3} + \frac{2}{5 \cdot 9} - \cdots$$

Homework

1. A ball is dropped from a meters above a flat surface. Each time the ball hits the surface after falling a distance h, it rebounds a distance rh, where r is a positive constant that is less than 1. Find the total distance the ball travels up and down.
2. Express the repeating decimal $5.232323\cdots$ as the ratio of two integers.
3. Find the sum of the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{n(n+1)} + \cdots .$$

4. Show that the series $\displaystyle\sum_{n=1}^{\infty} n^2$ diverges.
5. Show that the series $\displaystyle\sum_{n=1}^{\infty} \frac{n+1}{n}$ diverges.

Problem:

Find the sum of the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{n(n+1)} + \cdots$$

Solution: Here the nth term $\dfrac{1}{n(n+1)}$ can be written as

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1} \quad (n = 1, 2, 3, \ldots).$$

So, the nth partial sum can be written as a telescoping sum:

$$\begin{aligned} s_n &= \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \cdots + \frac{1}{n(n+1)} \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\ &= 1 - \frac{1}{n+1}. \end{aligned}$$

Thus $s_n \to 1$ as $n \to \infty$. Hence the sum of the series is 1.

Diverging Series

Show that the series $\sum_{n=1}^{\infty} n^2$ diverges.

Solution: Here the nth partial sum is

$$s_n = 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Obviously, $s_n \to \infty$ as $n \to \infty$. Hence the series diverges.

Diverging Series

Show that the series $\sum_{n=1}^{\infty} \frac{n+1}{n}$ diverges.

Solution: The nth partial sum

$$\begin{aligned} s_n &= 2 + \frac{3}{2} + \frac{4}{3} + \cdots + \frac{n+1}{n} \\ &> 1 + 1 + 1 + \cdots + 1 \\ &= n. \end{aligned}$$

So, the sequence of partial sums $\{s_n\}$ is unbounded. This implies the sequence $\{s_n\}$ is divergent. Hence, the series diverges.

An Important Theorem

Theorem

If $\sum_{n=1}^{\infty} a_n$ *converges, then* $a_n \to 0$.

Proof.

- Let $\sum_{n=1}^{\infty} a_n = \ell$.
- This means that the sequence $\{s_n\}$ of partial sums converges to ℓ.
- But $a_n = s_n - s_{n-1}$ and $s_n \to \ell$ and $s_{n-1} \to \ell$ as $n \to \infty$.
- Thus $a_n = s_n - s_{n-1} \to \ell - \ell = 0$ as $n \to \infty$.

□

The nth Term Test for Divergence

If $\lim_{n\to\infty} a_n$ does not exist or is different from 0, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

Applying the nth Term Test for Divergence

1. $\sum_{n=1}^{\infty} n^2$ diverges, because $n^2 \to \infty$ as $n \to \infty$.
2. $\sum_{n=1}^{\infty} \frac{n+1}{n}$ diverges, because $\frac{n+1}{n} \to 1 \neq 0$ as $n \to \infty$.
3. $\sum_{n=1}^{\infty} (-1)^{n+1}$ diverges, because $\lim_{n\to\infty} (-1)^{n+1}$ does not exist.
4. $\sum_{n=1}^{\infty} \frac{-n}{2n+5}$ diverges, because $\frac{-n}{2n+5} \to -\frac{1}{2} \neq 0$ as $n \to \infty$.

The Converse of the Theorem is not True

$a_n \to 0$ does not imply that $\sum_{n=1}^{\infty} a_n$ converges.

Example: Consider the series $\sum_{n=1}^{\infty} \frac{1}{n}$.

It's nth term $\frac{1}{n} \to 0$ as $n \to \infty$, but the series diverges.

Note

We will often denote the series $\sum_{n=1}^{\infty} a_n$ simply as $\sum a_n$.

Combining Series

Theorem

Let $\sum a_n = a$ *and* $\sum b_n = b$. *Then*

1. $\sum (a_n + b_n) = a + b.$
2. $\sum (a_n - b_n) = a - b.$
3. $\sum (ka_n) = ka$ *for any number* k.

Proof (1):

- Let $A_n = a_1 + a_2 + a_3 + \cdots + a_n$ and $B_n = b_1 + b_2 + b_3 + \cdots + b_n$.
- Then the partial sums of the series $\sum(a_n + b_n)$ are

$$\begin{aligned} s_n &= (a_1 + b_1) + (a_2 + b_2) + \cdots + (a_n + b_n) \\ &= (a_1 + a_2 + a_3 + \cdots + a_n) + (b_1 + b_2 + b_3 + \ldots + b_n) \\ &= A_n + B_n. \end{aligned}$$

- But $A_n \to a$ and $B_n \to b$.
- Hence $s_n = A_n + B_n \to a + b$ by the Addition Rule for sequences.

Corollary

1. *If* $\sum a_n$ *diverges and* c *is a non-zero constant, then* $\sum ca_n$ *also diverges.*
2. *If* $\sum a_n$ *converges and* $\sum b_n$ *diverges, then* $\sum(a_n + b_n)$ *and* $\sum(a_n - b_n)$ *both diverge.*

Example

Find the sum of the series

$$\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}}.$$

Solution:

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{3^{n-1}-1}{6^{n-1}} &= \sum_{n=1}^{\infty} \left(\frac{1}{2^{n-1}} - \frac{1}{6^{n-1}} \right) \\
&= \sum_{n=1}^{\infty} \frac{1}{2^{n-1}} - \sum_{n=1}^{\infty} \frac{1}{6^{n-1}} \\
&= \frac{1}{1-(1/2)} - \frac{1}{1-(1/6)} \\
&= 2 - \frac{6}{5} \\
&= \frac{4}{5}.
\end{aligned}$$

Note

Addition or deletion of a finite number of terms does not affect the convergence or divergence of a series. But in the case of convergent series, this may change the sum of the series.

Example:

$$1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \cdots$$

An Important Series

Show that the series

$$\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} + \cdots$$

converges.

Solution: Here the nth partial sum is

$$\begin{aligned} s_n &= 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} \\ &< 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}} \\ &< 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}} + \cdots \\ &= 1 + 2 \\ &= 3. \end{aligned}$$

The sequence of partial sums is bounded above by 3. It is also monotonically increasing. Thus, the series $\sum_{n=0}^{\infty} \frac{1}{n!}$ converges.

An Important Series

Definition

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} + \cdots .$$

Note: $e = 2.7182\ldots$

Tests of Convergence of Series: The Comparison Test

Theorem (The Comparison Test)

Let $\sum a_n$ be a series of non-negative terms. Then

(a) *$\sum a_n$ converges if there is a convergent series $\sum c_n$ with $a_n \leq c_n$ for all $n \geq N$, for some integer N.*

(b) *$\sum a_n$ diverges if there is a divergent series of non-negative terms $\sum d_n$ with $a_n \geq d_n$ for all $n \geq N$, for some integer N.*

Proof of (a): The sequence of partial sums $\{s_n\}$ of $\sum a_n$ is a monotonically increasing sequence and is bounded above by

$$M = a_1 + a_2 + \cdots + a_{N-1} + \sum_{n=N}^{\infty} c_n.$$

So it converges, and hence the series $\sum a_n$ converges.

Proof of (b):

Given that $\sum d_n$ diverges.

Assume that the series $\sum a_n$ converges. As d_n are non-negative and $d_n \leq a_n$ for $n \geq N$, by part (a) the series $\sum d_n$ converges. This is a contradiction to our assumption that $\sum a_n$ converges. Hence the series $\sum a_n$ diverges.

Example: The Harmonic Series

The series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots$$

is called the **harmonic series**.

The harmonic series diverges

Note that

$$1+\frac{1}{2}+\left(\frac{1}{3}+\frac{1}{4}\right)+\left(\frac{1}{5}+\frac{1}{6}+\frac{1}{7}+\frac{1}{8}\right)+\left(\frac{1}{9}+\frac{1}{10}+\cdots+\frac{1}{16}\right)+\cdots$$

$$>1+\frac{1}{2}+\left(\frac{1}{4}+\frac{1}{4}\right)+\left(\frac{1}{8}+\frac{1}{8}+\frac{1}{8}+\frac{1}{8}\right)+\left(\frac{1}{16}+\cdots+\frac{1}{16}\right)+\cdots$$

$$>1+\frac{1}{2}+2\cdot\frac{1}{2^2}+2^2\cdot\frac{1}{2^3}+2^3\cdot\frac{1}{2^4}+\cdots$$

$$=1+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\cdots$$

As the series $1+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\frac{1}{2}+\cdots$ diverges, by comparison test the harmonic series $1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots$ diverges.

Examples

1. The series
$$\sum_{n=1}^{\infty} \frac{7}{7n-2}$$
diverges, because its nth term
$$\frac{7}{7n-2} = \frac{1}{n-\frac{2}{7}} > \frac{1}{n}$$
which is the nth term of the divergent harmonic series.

2. The series
$$1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!} + \ldots$$
converges because its terms are all positive and are less than or equal to the corresponding terms of
$$1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}} + \cdots$$
which is convergent.

Tests of Convergence of Series: The Limit Comparison Test

Theorem (The Limit Comparison Test)

Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$ (N an integer).

1. *If* $\lim_{n\to\infty} \frac{a_n}{b_n} = c > 0$, *then* $\sum a_n$ *and* $\sum b_n$ *both converge or both diverge.*
2. *If* $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ *and* $\sum b_n$ *converges, then* $\sum a_n$ *converges.*
3. *If* $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$ *and* $\sum b_n$ *diverges, then* $\sum a_n$ *diverges.*

Example

Does the following series converge?

$$\sum_{n=1}^{\infty} \frac{2n+1}{(n+1)^2}.$$

Solution: Let $a_n = \frac{2n+1}{(n+1)^2} = \frac{2n+1}{n^2+2n+1}$. For n large, we expect a_n to behave like $2n/n^2 = 2/n$. So, we let $b_n = 1/n$.
Since

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n}$$

diverges and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{2n^2+n}{n^2+2n+1} = 2,$$

$\sum a_n$ diverges by Part 1 of the Limit Comparison Test.

Example

Does the following series converge or diverge?

$$\sum_{n=2}^{\infty} \frac{1 + n \ln n}{n^2 + 5}.$$

Solution: Let $a_n = (1 + n \ln n)/(n^2 + 5)$. For large n, a_n will behave like $n \ln n / n^2 = \ln n / n$, which is greater than $1/n$ for $n \geq 3$. So, we take $b_n = 1/n$.
Since

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n}$$

diverges and

$$\lim_{n \to \infty} \frac{a_n}{b_n} = \lim_{n \to \infty} \frac{n + n^2 \ln n}{n^2 + 5} = \lim_{n \to \infty} \frac{\frac{1}{n} + \ln n}{1 + \frac{5}{n^2}} = \infty,$$

$\sum a_n$ diverges by Part 3 of the Limit Comparison Test.

Homework

Does $\sum_{n=1}^{\infty} \frac{\ln n}{n^{3/2}}$ converge?

Tests of Convergence of Series: The Limit Comparison Test

Theorem (The Limit Comparison Test)

Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$ (N an integer).

1. *If* $\lim_{n\to\infty} \frac{a_n}{b_n} = c > 0$, *then* $\sum a_n$ *and* $\sum b_n$ *both converge or both diverge.*
2. *If* $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ *and* $\sum b_n$ *converges, then* $\sum a_n$ *converges.*
3. *If* $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$ *and* $\sum b_n$ *diverges, then* $\sum a_n$ *diverges.*

Proof of (1):

Since $\frac{c}{2} > 0$, there exists an integer N such that

$$n \geq N \quad \Rightarrow \quad \left| \frac{a_n}{b_n} - c \right| < \frac{c}{2}.$$

Thus, for $n \geq N$,

$$-\frac{c}{2} < \frac{a_n}{b_n} - c < \frac{c}{2},$$

$$\Leftrightarrow \frac{c}{2} < \frac{a_n}{b_n} < \frac{3c}{2},$$

$$\Leftrightarrow \left(\frac{c}{2}\right) b_n < a_n < \left(\frac{3c}{2}\right) b_n.$$

Proof of (1):

If $\sum b_n$ converges, then $\sum(3c/2)b_n$ converges and $\sum a_n$ converges by the direct Comparison Test.

If $\sum b_n$ diverges, then $\sum(c/2)b_n$ diverges and $\sum a_n$ diverges by the direct Comparison Test.

Similarly, if $\sum a_n$ converges, then $\sum(c/2)b_n$ converges by the direct comparison test, and hence $\sum b_n$ converges.

If $\sum a_n$ diverges, then $\sum(3c/2)b_n$ diverges by the direct comparison test, and hence $\sum b_n$ diverges.

Proof of (2):

Given that $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$. Let $\epsilon > 0$. Then there exists an integer N such that

$$\left|\frac{a_n}{b_n}\right| < \epsilon \quad \text{for } n \geq N.$$

Thus, for $n \geq N$,

$$-\epsilon < \frac{a_n}{b_n} < \epsilon,$$

$$\Leftrightarrow -\epsilon b_n < a_n < \epsilon b_n.$$

If $\sum b_n$ converges, then $\sum \epsilon b_n$ converges and hence $\sum a_n$ converges by the direct Comparison Test.

Proof of (3):

Given that $\lim_{n\to\infty} \frac{a_n}{b_n} = \infty$. Let $M > 0$. Then there exists an integer N such that

$$\left|\frac{a_n}{b_n}\right| > M \quad \text{for } n \geq N.$$

Thus, for $n \geq N$,

$$\frac{a_n}{b_n} > M,$$
$$\Leftrightarrow a_n > Mb_n.$$

If $\sum b_n$ diverges, then $\sum Mb_n$ diverges and hence $\sum a_n$ diverges by the direct Comparison Test.

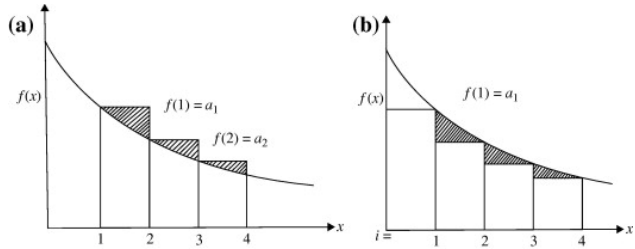
Tests of Convergence of Series: The Integral Test

Theorem (The Integral Test)

Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, decreasing function of x for all $n \geq N$ (N a positive integer). Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_{n=N}^{\infty} f(x)dx$ both converge or both diverge.

Proof

We will prove the theorem for $N = 1$. Let f be a function with $f(n) = a_n$ for all n.

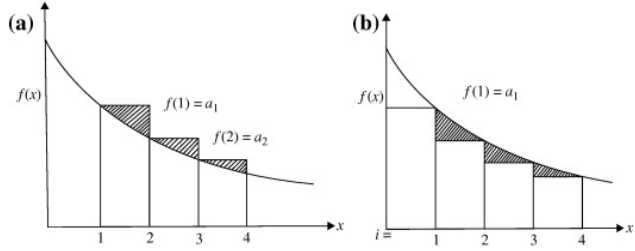


The rectangles in Figure (a) which have areas $a_1, a_2, \ldots, a_n$ collectively enclose an area $a_1 + a_2 + \ldots + a_n$. This is more than the area under the curve $y = f(x)$ from 1 to $n + 1$. That is,

$$\int_1^{n+1} f(x)dx \leq a_1 + a_2 + \ldots + a_n.$$

Note: The image is from the internet.

Proof:



The rectangles in Figure (b) having areas $a_2, a_3, \ldots, a_n$ collectively enclose an area $a_2 + a_3 + \ldots + a_n$. This is less than the area under the curve $y = f(x)$ from 1 to n. That is,

$$a_2 + a_3 + \ldots + a_n \leq \int_1^n f(x)dx.$$

Adding a_1 to both sides, we get

$$a_1 + a_2 + a_3 + \ldots + a_n \leq a_1 + \int_1^n f(x)dx.$$

Combining the two inequalities gives

$$\int_1^{n+1} f(x)dx \leq a_1 + a_2 + \ldots + a_n \leq a_1 + \int_1^n f(x)dx.$$

The above inequalities hold for each n and continue to hold as $n \to \infty$.

If $\int_1^\infty f(x)dx$ is finite, the right-hand inequality implies that $\sum a_n$ is finite.

If $\int_1^\infty f(x)dx$ is infinite, the left-hand inequality implies that $\sum a_n$ is infinite.

Hence the series and the integral are both finite or both infinite.

The p-Series

The series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \cdots$$

is called the p-series (p any real constant).

Example

Show that the p-series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \cdots + \frac{1}{n^p} + \cdots$$

converges if $p > 1$ and diverges if $p \leq 1$.

Solution: If $p > 0$, then $f(x) = \frac{1}{x^p}$ is a positive, decreasing, and continuous function of x for $x \geq 1$. Also, $f(n) = \frac{1}{n^p}$ for $n \geq 1$. If $p > 1$, then

$$\begin{aligned}
\int_1^{\infty} \frac{1}{x^p} dx &= \int_1^{\infty} x^{-p} dx \\
&= \left[\frac{x^{-p+1}}{-p+1} \right]_1^{\infty} \\
&= \frac{1}{1-p} \left[\frac{1}{x^{p-1}} \right]_1^{\infty} \\
&= \frac{1}{1-p}[0-1] = \frac{1}{p-1}.
\end{aligned}$$

So, the series converges by the integral test in this case.

If $0 < p < 1$, then $1 - p > 0$ and

$$\int_1^\infty \frac{1}{x^p} dx = \int_1^\infty x^{-p} dx = \left[\frac{x^{-p+1}}{-p+1} \right]_1^\infty = \frac{1}{1-p} \left[x^{1-p} \right]_1^\infty = \infty.$$

If $p = 1$, we have the (divergent) harmonic series:

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots$$

Thus the p-series converges for $p > 1$ and diverges for $0 < p \leq 1$.

Also the p-series diverges if $p \leq 0$. (Why?)

Homework

1. Show that the series

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1}$$

converges using the integral test.

2. Show that the series

$$\sum_{n=1}^{\infty} \frac{1}{n(\log n)^p}$$

converges if and only if $p > 1$ using the integral test.

The Ratio Test

Theorem (The Ratio Test)

Let $\sum a_n$ be a series with positive terms and suppose that

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \rho.$$

Then

(a) *the series* converges *if* $\rho < 1$,

(b) *the series* diverges *if* $\rho > 1$ *or is infinite,*

(c) *the test is inconclusive if* $\rho = 1$.

Example

Investigate the convergence of the series

$$\sum_{n=1}^{\infty} \frac{2^n + 5}{3^n}.$$

Solution: Here

$$\frac{a_{n+1}}{a_n} = \frac{(2^{n+1} + 5)/3^{n+1}}{(2^n + 5)/3^n} = \frac{1}{3} \cdot \frac{2^{n+1} + 5}{2^n + 5} \to \frac{2}{3}.$$

The series converges because here $\rho = 2/3 < 1$.

Example

Test the convergence of the series

$$\sum_{n=1}^{\infty} \frac{2^n}{n!}.$$

Solution: For the series $\sum_{n=1}^{\infty} \frac{2^n}{n!}$,

$$\frac{a_{n+1}}{a_n} = \frac{2^{n+1}/(n+1)!}{2^n/n!} = \frac{2}{n+1} \to 0.$$

The series converges because here $\rho = 0 < 1$.

Example

Test the convergence of the series

$$\sum_{n=1}^{\infty} \frac{4^n n! n!}{(2n)!}.$$

Solution: Here

$$\frac{a_{n+1}}{a_n} = \frac{4^{n+1}(n+1)!(n+1)!}{(2n+2)!} \cdot \frac{(2n)!}{4^n n! n!} = \frac{4(n+1)(n+1)}{(2n+2)(2n+1)} = \frac{2(n+1)}{2n+1} \to 1.$$

Because the limit $\rho = 1$, we cannot decide whether the series converges from the ratio test.

But

$$\frac{a_{n+1}}{a_n} = \frac{2(n+1)}{2n+1} > 1 \quad \text{for all} \quad n.$$

Thus $a_{n+1} > a_n$ for all n. That is $a_1 < a_2 < a_3 < \cdots$.
Also $a_1 = 2$. Thus a_n does not converge to 0. Hence the series diverges.

The Ratio Test

Theorem (The Ratio Test)

Let $\sum a_n$ be a series with positive terms and suppose that

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \rho.$$

Then

(a) *the series* converges *if* $\rho < 1$,

(b) *the series* diverges *if* $\rho > 1$ *or is infinite,*

(c) *the test is inconclusive if* $\rho = 1$.

Proof.

(a) $\rho < 1$.

Let r be a number between ρ and 1. That is $\rho < r < 1$.
Then the number $\epsilon = r - \rho$ is positive.
Since

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \rho,$$

there is an integer N such that

$$n \geq N \quad \Rightarrow \quad \left| \frac{a_{n+1}}{a_n} - \rho \right| < \epsilon.$$

This implies that, for $n \geq N$,

$$-\epsilon < \frac{a_{n+1}}{a_n} - \rho < \epsilon$$

$$\Leftrightarrow \rho - \epsilon < \frac{a_{n+1}}{a_n} < \rho + \epsilon = r.$$

That is,

$$\begin{aligned} a_{N+1} &< ra_N, \\ a_{N+2} &< ra_{N+1} < r^2 a_N, \\ a_{N+3} &< ra_{N+2} < r^3 a_N, \\ &\vdots \end{aligned}$$

Consider the series $\sum c_n$, where $c_n = a_n$ for $n = 1, 2, \ldots, N$ and $c_{N+1} = ra_N$, $c_{N+2} = r^2 a_N, \ldots$
Now $a_n \leq c_n$ for all n and

$$\begin{aligned} \sum_{n=1}^{\infty} c_n &= a_1 + a_2 + \cdots + a_{N-1} + a_N + ra_N + r^2 a_N + \cdots \\ &= a_1 + a_2 + \cdots + a_{N-1} + a_N(1 + r + r^2 + \cdots). \end{aligned}$$

The geometric series $1 + r + r^2 + \cdots$ converges as $r < 1$. So $\sum c_n$ converges.
Since $a_n \leq c_n$, $\sum a_n$ also converges by the direct comparison test.

(b) $1 < \rho \leq \infty$.

Choose ϵ to be a positive real number sch that $\epsilon < \rho - 1$. For $\epsilon > 0$, there exists a natural number N such that

$$\left| \frac{a_{n+1}}{a_n} - \rho \right| < \epsilon$$

$$\Leftrightarrow -\epsilon < \frac{a_{n+1}}{a_n} - \rho < \epsilon$$

$$\Leftrightarrow \frac{a_{n+1}}{a_n} > \rho - \epsilon > 1.$$

Thus, from index N on

$$\frac{a_{n+1}}{a_n} > 1.$$

That is

$$a_N < a_{N+1} < a_{N+2} < \cdots .$$

So, the terms of the series do not approach zero as n becomes infinite. Hence the series diverges by the nth-term Test.

(c) $\rho = 1$.

Consider the following two series:

$$\sum_{n=1}^{\infty} \frac{1}{n} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

For $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$: $\displaystyle\frac{a_{n+1}}{a_n} = \frac{1/(n+1)}{1/n} = \frac{n}{n+1} \to 1.$

For $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$: $\displaystyle\frac{a_{n+1}}{a_n} = \frac{1/(n+1)^2}{1/n^2} = \left(\frac{n}{n+1}\right)^2 \to 1.$

In both cases, $\rho = 1$. But the first series diverges and the second converges.

The Root Test

Theorem (The Root Test)

Let $\sum a_n$ be a series with $a_n \geq 0$ for $n \geq N$ and suppose that

$$\lim_{n\to\infty} \sqrt[n]{a_n} = \rho.$$

Then

(a) *the series converges if $\rho < 1$,*

(b) *the series diverges if $\rho > 1$ or ρ is infinite,*

(c) *the test is inconclusive if $\rho = 1$.*

Proof.

(a) $\rho < 1$.

Let r be a number between ρ and 1. That is $\rho < r < 1$.
Then the number $\epsilon = r - \rho$ is positive.
Since

$$\lim_{n\to\infty} a_n^{1/n} = \rho,$$

there is an integer N such that

$$n \geq N \quad \Rightarrow \quad \left| a_n^{1/n} - \rho \right| < \epsilon.$$

This implies that, for $n \geq N$,

$$\rho - \epsilon < a_n^{1/n} < \rho + \epsilon = r.$$

$$\Rightarrow a_n < r^n.$$

That is,

$$\begin{aligned} a_N &< r^N, \\ a_{N+1} &< r^{N+1}, \\ a_{N+2} &< r^{N+2}, \\ &\vdots \end{aligned}$$

Consider the series $\sum c_n$, where $c_n = a_n$ for $n = 1, 2, \ldots, N-1$ and $c_N = r^N, c_{N+1} = r^{N+1}, c_{N+2} = r^{N+2}, \ldots$

Now $a_n \leq c_n$ for all n and

$$\begin{aligned} \sum_{n=1}^{\infty} c_n &= a_1 + a_2 + \cdots + a_{N-1} + r^N + r^{N+1} + r^{N+2} + \cdots \\ &= a_1 + a_2 + \cdots + a_{N-1} + r^N(1 + r + r^2 + \cdots). \end{aligned}$$

The geometric series $1 + r + r^2 + \cdots$ converges as $r < 1$. So, $\sum c_n$ converges.
Since $a_n \leq c_n$, $\sum a_n$ also converges by the direct comparison test.

(b) $1 < \rho \leq \infty$.

Choose ϵ to be a positive real number sch that $\epsilon < \rho - 1$. For $\epsilon > 0$, there exists a natural number N such that

$$\left| a_n^{1/n} - \rho \right| < \epsilon$$

$$\Leftrightarrow \rho - \epsilon < a_n^{1/n} < \rho + \epsilon$$

$$\Rightarrow a_n^{1/n} > \rho - \epsilon > 1.$$

Thus, from index N on

$$a_n^{1/n} > 1.$$

Equivalently, for $n \geq N$,

$$a_n > 1.$$

So, the terms of the series do not approach zero as n becomes infinite. Hence the series diverges by the nth-term Test.

(c) $\rho = 1$.

Consider the following two series:

$$\sum_{n=1}^{\infty} \frac{1}{n} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

For the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$, $\quad a_n^{1/n} = \frac{1}{n^{1/n}} \to 1$ as $n \to \infty$.

For the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$, $\quad a_n^{1/n} = \left(\frac{1}{n^{1/n}}\right)^2 \to 1$ as $n \to \infty$.

In both cases, $\rho = 1$. But the first series diverges and the second converges.

Examples

Which of the following series converges and which diverges?

$$\text{(a)} \sum_{n=1}^{\infty} \frac{n^2}{2^n} \qquad \text{(b)} \sum_{n=1}^{\infty} \frac{2^n}{n^2} \qquad \text{(c)} \sum_{n=1}^{\infty} \left(\frac{1}{1+n}\right)^n$$

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ converges because $\displaystyle\sqrt[n]{\frac{n^2}{2^n}} = \frac{(\sqrt[n]{n})^2}{2} \to \frac{1}{2} < 1.$

(b) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{2^n}$ diverges because $\displaystyle\sqrt[n]{\frac{2^n}{n^2}} = \frac{2}{(\sqrt[n]{n})^2} \to \frac{2}{1} > 1.$

(c) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{1}{1+n}\right)^n$ converges because $\displaystyle\sqrt[n]{\left(\frac{1}{1+n}\right)^n} = \frac{1}{1+n} \to 0 < 1.$

Example

Let $a_n = \begin{cases} n/2^n, & n \text{ odd} \\ 1/2^n, & n \text{ even.} \end{cases}$ Does $\sum a_n$ converge?

Solution: Here

$$\sqrt[n]{a_n} = \begin{cases} \sqrt[n]{n}/2, & n \text{ odd} \\ 1/2, & n \text{ even.} \end{cases}$$

Therefore,

$$\frac{1}{2} \leq \sqrt[n]{a_n} \leq \frac{\sqrt[n]{n}}{2}.$$

We know that $\sqrt[n]{n} \to 1$. So, $\lim_{n\to\infty} \sqrt[n]{a_n} = 1/2$ by the Sandwich Theorem.

Thus here the limit is $\rho < 1$. Hence the series converges by the Root Test.

Alternating Series

Definition

A series in which the terms are alternatively positive and negative is called an **alternating series**.

Examples

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

$$-2 + 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots = \sum_{n=1}^{\infty} \frac{(-1)^{n} 4}{2^{n}}$$

$$1 - 2 + 3 - 4 + 5 - 6 + \cdots = \sum_{n=1}^{\infty} (-1)^{n+1} n$$

The first series, called the **alternating harmonic series**, converges.

The second series, a geometric series with common ratio $r = -1/2$, converges.

The third series diverges because the nth term does not approach zero.

The Alternating Series Test

Theorem (The Alternating Series Test (Leibniz's Theorem))

The series

$$\sum_{n=1}^{\infty}(-1)^{n+1}u_n = u_1 - u_2 + u_3 - u_4 + \cdots$$

converges if all three of the following conditions are satisfied:

1. *The u_n's are all positive.*
2. *$u_n \geq u_{n+1}$ for all $n \geq N$, for some integer N.*
3. *$u_n \to 0$.*

Example

The alternating harmonic series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

satisfies the three requirements of the alternating series theorem with $N = 1$. So, it converges.

Homework

Let $\{a_n\}$ be a sequence such that the subsequences $\{a_{2m}\}$ and $\{a_{2m+1}\}$ both converge to the same limit ℓ. Then show that $a_n \to \ell$.

Proof of Leibinz's Theorem.

Let us assume that $N = 1$.
If n is an even integer, say $n = 2m$, then the sum of the first n terms is

$$\begin{aligned} s_{2m} &= (u_1 - u_2) + (u_3 - u_4) + \cdots + (u_{2m-1} - u_{2m}) \\ &= u_1 - (u_2 - u_3) - (u_4 - u_5) - \cdots - (u_{2m-2} - u_{2m-1}) - u_{2m}. \end{aligned}$$

The first equality shows that the s_{2m} is the sum of m non-negative terms. Hence

$$s_{2m+2} \geq s_{2m}.$$

The second equality shows that

$$s_{2m} \leq u_1.$$

So, $\{s_{2m}\}$ is monotonically increasing and bounded above. So, it converges, say

$$\lim_{m \to \infty} s_{2m} = \ell.$$

Also $u_{2m+1} \to 0$. Hence

$$s_{2m+1} = s_{2m} + u_{2m+1} \to \ell + 0 = \ell.$$

Thus we have that

$$s_{2m} \to \ell \quad \text{and } s_{2m+1} \to \ell.$$

Hence

$$s_n \to \ell.$$

This means that the alternating series converges.

Absolute Convergence

Definition (Absolute Convergence)

A series $\sum a_n$ **converges absolutely** if the corresponding series of absolute values $\sum |a_n|$ converges.

Example 1. The geometric series

$$1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$$

converges absolutely because the corresponding series of absolute values

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$$

converges.

Example 2. In contrast, the alternating harmonic series does not converge absolutely: The corresponding series of absolute values is the divergent harmonic series.

The Absolute Convergence Test

Theorem

If $\sum_{n=1}^{\infty} |a_n|$ *converges, then* $\sum_{n=1}^{\infty} a_n$ *converges.*

Proof:

For each n, we have

$$-|a_n| \leq a_n \leq |a_n|, \qquad \text{so} \qquad 0 \leq a_n + |a_n| \leq 2|a_n|.$$

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} 2|a_n|$ converges.

So, by the Comparison Test, $\sum_{n=1}^{\infty} (a_n + |a_n|)$ converges.

But $a_n = (a_n + |a_n|) - |a_n|$. So, $\sum_{n=1}^{\infty} a_n$ can be expressed as the difference of two convergent series:

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (a_n + |a_n| - |a_n|) = \sum_{n=1}^{\infty} (a_n + |a_n|) - \sum_{n=1}^{\infty} |a_n|.$$

Therefore, $\sum_{n=1}^{\infty} a_n$ converges.

Examples

(a) Consider the series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^3} = 1 - \frac{1}{8} + \frac{1}{27} - \frac{1}{64} + \cdots$$

The corresponding series of absolute values is

$$\sum_{n=1}^{\infty} \frac{1}{n^3} = 1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} + \cdots$$

The latter converges. Thus the original series converges absolutely. Hence it converges.

Examples

(b) Consider the series

$$\sum_{n=1}^{\infty} \frac{\sin n}{n^2} = \frac{\sin 1}{1} + \frac{\sin 2}{4} + \frac{\sin 3}{9} + \cdots$$

The corresponding series of absolute values is

$$\sum_{n=1}^{\infty} \left| \frac{\sin n}{n^2} \right| = \left| \frac{\sin 1}{1} \right| + \left| \frac{\sin 2}{4} \right| + \left| \frac{\sin 3}{9} \right| + \cdots .$$

Since $|\sin n| \leq 1$, the latter converges by comparison with the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$.

Thus the original series converges absolutely. Hence it converges.

Definition (Conditional Convergence)

A series that converges but does not converge absolutely is said to **converge conditionally**.

Example: The alternating harmonic series converges conditionally.

Power Series

Definition (Power Series, Center, Coefficients)

A **power series about** $x = 0$ is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots .$$

A **power series about** $x = a$ is a series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + \cdots + c_n (x-a)^n + \cdots .$$

Here a is the **center** and $c_0, c_1, c_2, \ldots, c_n, \ldots$ are the **coefficients** of the power series. These are constants.

Example: A geometric series

Consider the power series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots + x^n + \cdots .$$

This the geometric series with first term 1 and common ratio x.
It converges to $1/(1-x)$ for $|x| < 1$. We express this fact by writing

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \cdots , \qquad -1 < x < 1.$$

Example: Another geometric series

The power series

$$1 - \frac{1}{2}(x-2) + \frac{1}{4}(x-2)^2 + \ldots + \left(-\frac{1}{2}\right)^n (x-2)^n + \cdots$$

has center $a = 2$ and coefficients $c_0 = 1, c_1 = -1/2, c_2 = 1/4, \ldots, c_n = (-1/2)^n, \ldots$.

This is a geometric series with first term 1 and common ratio $r = -\dfrac{x-2}{2}$. The series converges for $|r| = \left|-\dfrac{x-2}{2}\right| < 1$ or $0 < x < 4$.

The sum of the series is

$$\frac{1}{1-r} = \frac{1}{1+\frac{x-2}{2}} = \frac{2}{x}.$$

So,

$$\frac{2}{x} = 1 - \frac{1}{2}(x-2) + \frac{1}{4}(x-2)^2 + \cdots + \left(-\frac{1}{2}\right)^n (x-2)^n + \cdots, \quad 0 < x < 4.$$

Testing Power Series for Convergence Using the Ratio Test

Problem. For what values of x the following series converge?

$$\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots .$$

Solution: We apply the Ratio Test to the series $\sum_{n=1}^{\infty}|u_n|$, where u_n is the nth term of the given power series:

$$\left|\frac{u_{n+1}}{u_n}\right| = \left|\frac{x^{n+1}}{n+1}\frac{n}{x^n}\right| = \frac{n}{n+1}|x| \to |x|.$$

Thus the given power series converges absolutely for $|x| < 1$. It diverges if $|x| > 1$, because the n-th term of the series does not converge to zero.
At $x = 1$, it becomes the alternating harmonic series. So, it converges (conditionally).
At $x = -1$, we get the negative of the harmonic series. So, it diverges.

Testing Power Series for Convergence Using the Ratio Test

Summary: The series

$$\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$$

converges for $-1 < x \leq 1$ and diverges for other values of x.

Testing Power Series for Convergence Using the Ratio Test

Problem. For what values of x the following series converge?

$$\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^{2n-1}}{2n-1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots .$$

Solution: We apply the Ratio Test to the series $\sum_{n=1}^{\infty}|u_n|$, where u_n is the nth term of the given power series:

$$\left|\frac{u_{n+1}}{u_n}\right| = \left|\frac{x^{2n+1}}{2n+1}\frac{2n-1}{x^{2n-1}}\right| = \frac{2n-1}{2n+1}x^2 \to x^2.$$

Thus the given power series converges absolutely for $x^2 < 1$. It diverges for $x^2 > 1$, because the n-th term of the series does not converge to zero.
At $x = 1$, the series becomes the alternating series $1 - 1/3 + 1/5 - 1/7 + \cdots$, which converges by the Alternating Series Test.
The value at $x = -1$ is the negative of the value at $x = 1$. So, it converges for $x = -1$ as well.

Testing Power Series for Convergence Using the Ratio Test

Summary: The series

$$\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^{2n-1}}{2n-1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots$$

converges for $|x| \leq 1$ and diverges otherwise.

Testing Power Series for Convergence Using the Ratio Test

Problem. For what values of x the following series converge?

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots .$$

Solution: We apply the Ratio Test to the series $\sum_{n=1}^{\infty} |u_n|$, where u_n is the nth term of the given power series:

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{x^{n+1}}{(n+1)!} \frac{n!}{x^n} \right| = \frac{|x|}{n+1} \to 0 \quad \text{for every } x.$$

So, the series converges absolutely for all values of x.

Testing Power Series for Convergence Using the Ratio Test

Problem. For what values of x the following series converge?

$$\sum_{n=0}^{\infty} n!x^n = 1 + x + 2!x^2 + 3!x^3 + \cdots .$$

Solution: We apply the Ratio Test to the series $\sum_{n=0}^{\infty} |u_n|$, where u_n is the nth term of the given power series:

$$\left|\frac{u_{n+1}}{u_n}\right| = \left|\frac{(n+1)!x^{n+1}}{n!x^n}\right| = (n+1)|x| \to \infty \quad \text{for every } x \neq 0.$$

So, the nth term of the series does not converge to zero for $x \neq 0$. Hence the series diverges for all values of x except $x = 0$.

Theorem (The Convergence Theorem for Power Series)

If the power series $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots$ *converges for* $x = c \neq 0$, *then it converges absolutely for all* x *with* $|x| < |c|$.

If the series diverges for $x = d$, *then it diverges for all* x *with* $|x| > |d|$.

Proof: Suppose the series $\sum_{n=0}^{\infty} a_n c^n$ converges.

Then by the n-th term test $a_n c^n \to 0$. Therefore, corresponding to $\epsilon = 1$, there is an integer N such that, for $n \geq N$,

$$|a_n c^n| < 1 \quad \text{or} \quad |a_n| < \frac{1}{|c|^n}.$$

Now take any x such that $|x| < |c|$ and consider the series

$$|a_0| + |a_1 x| + \cdots + |a_{N-1} x^{N-1}| + |a_N x^N| + |a_{N+1} x^{N+1}| + \cdots .$$

There are only a finite number of terms prior to $|a_N x^N|$ and so their sum is finite.

Starting from $|a_N x^N|$, the sum of the terms is less than

$$\left|\frac{x}{c}\right|^N + \left|\frac{x}{c}\right|^{N+1} + \left|\frac{x}{c}\right|^{N+2} + \cdots .$$

But the above series is a geometric series with common ratio less than 1, since $|x| < |c|$. So, it converges.

Thus it follows that the given power series converges absolutely for $|x| < |c|$.

To prove the second half of the theorem, we use the first half.

Suppose that the power series diverges at $x = d$ and converges at a point x_0 with $|x_0| > |d|$.

Then by taking $c = x_0$, we can conclude by the first half of the theorem that the power series converges at $x = d$, which is a contradiction.

Thus, it follows that if the power series diverges for $x = d$, then it diverges for all x with $|x| > |d|$.

Note

A similar result is also there for power series of the form $\sum_{n=0}^{\infty} a_n(x-a)^n$.

Take $y = x - a$, then the series takes the form $\sum_{n=0}^{\infty} a_n y^n$.

From the previous theorem, the series $\sum_{n=0}^{\infty} a_n y^n$ converges for $y = c \neq 0$ implies that it converges absolutely for $|y| < |c|$. This means that the power series $\sum_{n=0}^{\infty} a_n(x-a)^n$ converges for x with $|x - a| < |c|$.

Again, from the previous theorem, the series $\sum_{n=0}^{\infty} a_n y^n$ diverges for $y = d \neq 0$ implies that it diverges for $|y| > |c|$. This means that the power series $\sum_{n=0}^{\infty} a_n(x-a)^n$ diverges for x with $|x - a| > |d|$.

The Radius of Convergence

Theorem

The convergence of the series $\sum_{n=0}^{\infty} c_n(x-a)^n$ *is described by one of the following three possibilities:*

1. *There is a positive number* R *such that the series diverges for* x *with* $|x-a| > R$ *but converges absolutely for* x *with* $|x-a| < R$ *(i.e., for* $a-R < x < a+R$*). The series may or may not converge at either of the end points* $x = a-R$ *and* $x = a+R$*.*
2. *The series converges absolutely for every* x $(R = \infty)$*.*
3. *The series converges at* $x = a$ *and diverges for all* $x \neq a$ $(R = 0)$*.*

Proof

- We will prove the theorem for power series of the form $\sum_{n=0}^{\infty} c_n x^n$.
- If the series converges for every x, then we are in Case 2.
- If the series converges only at $x = 0$, then we are in Case 3.
- Otherwise there is a nonzero number d such that $\sum c_n d^n$ diverges and there is a nonzero number c such that $\sum c_n c^n$ converges.
- Let S denote the set of all numbers x such that $\sum c_n x^n$ converges.
- Then S is nonempty as it contains 0.
- Also, S contains all nonzero real numbers x with $|x| \leq p$ by the Convergence Theorem.

Proof...

- Also by the Convergence Theorem again, the power series diverges for all x with $|x| > |d|$. So, $S \subset [-d, d]$.
- Thus, S is a bounded set.
- Then by the Completeness Property of the real numbers, S has a least upper bound $R > 0$.
- If $|x| > R$, then $x \notin S$ and so $\sum c_n x^n$ diverges.
- If $|x| < R$, then there is a number $b \in S$ such that $|x| < b < R$. So, $\sum c_n b^n$ converges. Therefore $\sum c_n x^n$ converges absolutely by the Convergence Theorem.

How to find the radius of convergence

For the power series $\sum_{n=0}^{\infty} a_n x^n$, consider

$$\lim_{n\to\infty}\left|\frac{u_{n+1}}{u_n}\right| = \lim_{n\to\infty}\left|\frac{a_{n+1}x^{n+1}}{a_n x^n}\right| = |x|\cdot\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|.$$

From the ratio test we know that, if

$$|x|\cdot\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| < 1,$$

then the series $\sum_{n=0}^{\infty} a_n x^n$ converges.

Equivalently, if

$$|x| < \lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right|,$$

then the series $\sum_{n=0}^{\infty} a_n x^n$ converges.

Radius of convergence and Interval of convergence

Definition

The **radius of convergence** of the power series $\sum_{n=0}^{\infty} a_n x^n$ is

$$R = \lim_{n\to\infty} \left| \frac{a_n}{a_{n+1}} \right|.$$

Definition

The set of points where the power series $\sum_{n=0}^{\infty} a_n x^n$ converges is called the **interval of convergence** of the series $\sum_{n=0}^{\infty} a_n x^n$.

Problem. Find the radius and interval of convergence of

$$\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots .$$

Solution: Here $a_n = \dfrac{(-1)^{n-1}}{n}$. So, the radius of convergence is

$$R = \lim_{n\to\infty}\left|\frac{a_n}{a_{n+1}}\right| = \lim_{n\to\infty}\frac{1/n}{1/(n+1)} = \lim_{n\to\infty}\frac{n+1}{n} = 1.$$

Thus, the given power series converges absolutely for $|x| < 1$ and diverges for $|x| > 1$.

At $x = 1$, it becomes the alternating harmonic series. So, it converges.

At $x = -1$, we get the negative of the harmonic series. So, it diverges.

Thus, the interval of converge of the power series is $-1 < x \leq 1$.

Problem. Find the radius and interval of convergence of

$$1 - \frac{1}{2}(x-2) + \frac{1}{4}(x-2)^2 + \cdots = \sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n (x-2)^n.$$

Solution: Here $a_n = \left(-\frac{1}{2}\right)^n$. So, the radius of convergence is

$$R = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right| = \lim_{n\to\infty} \frac{(1/2)^n}{(1/2)^{n+1}} = 2.$$

Thus, the given power series converges absolutely for $|x-2| < 2$ and diverges for $|x-2| > 2$.

At $x - 2 = 2$, the series becomes $1 - 1 + 1 - 1 + \cdots = \sum_{n=0}^{\infty} (-1)^n$, thus it is divergent.

At $x - 2 = -2$, the series becomes $1 + 1 + 1 + \cdots$, thus it is divergent.

Hence, the interval of convergence is $|x-2| < 2$ or $0 < x < 4$.

Exercise-1. Find the radius and interval of convergence of

$$\sum_{n=1}^{\infty} \frac{(3x-2)^n}{n}.$$

Also find for what values of x the series converges (a) absolutely (b) conditionally?

Exercise-2. Find the radius and interval of convergence of

$$\sum_{n=0}^{\infty} (-1)^n (4x+1)^n.$$

Also find for what values of x the series converges (a) absolutely (b) conditionally?

Theorem (The term-by-term differentiation theorem)

If $\sum_{n=0}^{\infty} c_n(x-a)^n$ *converges for* $a-R<x<a+R$ *for some* $R>0$, *it defines a function*

$$f(x)=\sum_{n=0}^{\infty} c_n(x-a)^n, \qquad a-R<x<a+R.$$

This function f *has derivatives all orders inside the interval of convergence. We can obtain the derivatives by differentiating the original series term by term:*

$$\begin{aligned} f'(x) &= \sum_{n=1}^{\infty} nc_n(x-a)^{n-1}, \\ f''(x) &= \sum_{n=2}^{\infty} n(n-1)c_n(x-a)^{n-2}, \end{aligned}$$

and so on. Each of these derived series converges at every interior point of the interval of convergence of the original series.

Problem. Find the power series representations of $f'(x)$ and $f''(x)$ if $f(x) = \dfrac{1}{1-x}$.

Solution: We know that the power series representation of $f(x)$ is

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots, \quad -1 < x < 1.$$

So,

$$f'(x) = \frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + 4x^3 + \cdots + nx^{n-1} + \cdots, \quad -1 < x < 1.$$

Similarly,

$$f''(x) = \frac{2}{(1-x)^3} = 2 + 6x + 12x^2 + \cdots + n(n-1)x^{n-2} + \cdots, \quad -1 < x < 1.$$

Theorem (The term-by-term integration theorem)

Suppose that

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$$

converges for $a - R < x < a + R$ $(R > 0)$. *Then*

$$\sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$$

converges for $a - R < x < a + R$ *and*

$$\int f(x)dx = \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1} + C$$

for $a - R < x < a + R$.

Problem. Identify the function

$$f(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots, \qquad -1 \le x \le 1.$$

Solution: Differentiating the given series, we get

$$f'(x) = 1 - x^2 + x^4 - x^6 + \cdots, \qquad -1 < x < 1.$$

This is a geometric series with first term 1 and common ratio $-x^2$. So,

$$f'(x) = \frac{1}{1-(-x^2)} = \frac{1}{1+x^2}.$$

Now integration gives

$$f(x) = \int f'(x)dx = \int \frac{dx}{1+x^2} = \tan^{-1} x + C$$

The series for $f(x)$ is zero when $x = 0$; so $C = 0$. Hence

$$f(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots = \tan^{-1} x, \qquad -1 < x < 1.$$

Example: A series for $\ln(1+x)$

The series $1 - t + t^2 - t^3 + \cdots$ converges to the function $\dfrac{1}{1+t}$ for $-1 < t < 1$. Therefore,

$$\frac{1}{1+t} = 1 - t + t^2 - t^3 + \cdots \qquad -1 < t < 1.$$

$$\begin{aligned}
\ln(1+x) &= \int_0^x \frac{1}{1+t}\,dt \\
&= \int_0^x [1 - t + t^2 - t^3 + \cdots]\,dt \\
&= \left[t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \cdots\right]_0^x \\
&= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots, \qquad -1 < x < 1.
\end{aligned}$$

The Algebra of Power Series

- If $A(x) = \sum_{n=0}^{\infty} a_n x^n$ and $B(x) = \sum_{n=0}^{\infty} b_n x^n$ converge absolutely for $|x| < R$ $(R > 0)$, then $\sum_{n=0}^{\infty} (a_n \pm b_n) x^n$ converges absolutely to $A(x) \pm B(x)$ for $|x| < R$. That is

$$\sum_{n=0}^{\infty} a_n x^n \pm \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} (a_n \pm b_n) x^n.$$

- If $A(x) = \sum_{n=0}^{\infty} a_n x^n$ converge absolutely for $|x| < R$ $(R > 0)$ and k is a non-zero real number, then $\sum_{n=0}^{\infty} k a_n x^n$ converges absolutely to $kA(x)$ for $|x| < R$. That is

$$\sum_{n=0}^{\infty} k a_n x^n = k \sum_{n=0}^{\infty} a_n x^n.$$

The Algebra of Power Series

Exercise. Find the power series representation of the following functions

1. $\frac{1}{(1-x)(x-2)}$
2. $\frac{1}{(1-x)(x-3)}$.

Theorem (The multiplication theorem for power series)

If $A(x) = \sum_{n=0}^{\infty} a_n x^n$ *and* $B(x) = \sum_{n=0}^{\infty} b_n x^n$ *converge absolutely for* $|x| < R$ $(R > 0)$ *and*

$$c_n = a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \cdots + a_{n-1} b_1 + a_n b_0 = \sum_{k=0}^{n} a_k b_{n-k},$$

then $\sum_{n=0}^{\infty} c_n x^n$ *converges absolutely to* $A(x)B(x)$ *for* $|x| < R$*:*

$$\left(\sum_{n=0}^{\infty} a_n x^n \right) \cdot \left(\sum_{n=0}^{\infty} b_n x^n \right) = \sum_{n=0}^{\infty} c_n x^n.$$

Example

Multiply the geometric series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots = \frac{1}{1-x} \quad (|x| < 1)$$

by itself to get a power series for $\frac{1}{(1-x)^2}$ for $|x| < 1$.

Solution: Let

$$A(x) = \sum_{n=0}^{\infty} a_n x^n = 1 + x + x^2 + \ldots + x^n + \cdots = \frac{1}{1-x}$$

$$B(x) = \sum_{n=0}^{\infty} b_n x^n = 1 + x + x^2 + \cdots + x^n + \ldots = \frac{1}{1-x}$$

and

$$\begin{aligned} c_n &= c_n = a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \cdots + a_{n-1} b_1 + a_n b_0 \\ &= 1 + 1 + \cdots + 1 = n + 1 \end{aligned}$$

Then, by the Multiplication Theorem, the power series for $\dfrac{1}{(1-x)^2}$ is

$$\begin{aligned} A(x) \cdot B(x) &= \sum_{n=0}^{\infty} c_n x^n = \sum_{n=0}^{\infty} (n+1) x^n \\ &= 1 + 2x + 3x^2 + 4x^3 + \cdots, \quad |x| < 1. \end{aligned}$$

Taylor and Maclaurin Series

Suppose that

$$f(x) = \sum_{n=0}^{\infty} a_n(x-a)^n = a_0 + a_1(x-a) + a_2(x-a)^2 + \cdots + a_n(x-a)^n + \cdots$$

converges for $a - R < x < a + R$ $(R > 0)$.

Then the Term-by-Term Differentiation Theorem tells us that the sum function has derivatives of all orders within the interval of convergence $a - R < x < a + R$ and

$$\begin{aligned} f'(x) &= a_1 + 2a_2(x-a) + 3a_3(x-a)^2 + \cdots \\ f''(x) &= 1 \cdot 2a_2 + 2 \cdot 3a_3(x-a) + 3 \cdot 4a_4(x-a)^2 + \cdots \\ f'''(x) &= 1 \cdot 2 \cdot 3a_3 + 2 \cdot 3 \cdot 4a_4(x-a) + 3 \cdot 4 \cdot 5a_5(x-a)^2 + \cdots \end{aligned}$$

and so on. The above equations all hold, in particular, at $x = a$.

So,

$$\begin{aligned} f(a) &= a_0 \\ f'(a) &= a_1 \\ f''(a) &= 1 \cdot 2a_2 \\ f'''(a) &= 1 \cdot 2 \cdot 3a_3 \\ &\vdots \\ f^{(n)}(a) &= n!a_n. \end{aligned}$$

Thus

$$a_n = \frac{f^{(n)}(a)}{n!} \quad \text{for } n = 0, 1, 2, \ldots.$$

Hence

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots.$$

This implies that the sum function $f(x)$ has a unique power series expansion. (Why?)

Definition (Taylor Series, Maclaurin Series)

Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the **Taylor series generated by f at $x = a$** is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots .$$

The **Maclaurin series generated by f** is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + + \frac{f'''(0)}{3!}x^3 + \cdots ,$$

which is the Taylor series generated by f at $x = 0$.

Note: The Maclaurin series generated by f is often called the Taylor series of f.

Example

Problem. Find the Taylor series generated by $f(x) = 1/x$ at $a = 2$. Where, if anywhere, does the series converge to $f(x)$?

Solution: We need to find $f(2), f'(2), f''(2), \ldots$. Taking derivatives, we get

$$f(x) = x^{-1}, \quad f'(x) = -x^{-2}, \quad f''(x) = 2!x^{-3}, \quad f'''(x) = -3!x^{-4}, \ldots, f^{(n)}(x) = (-1)^n n! x^{-(n+1)}, \ldots.$$

So,

$$f(2) = \frac{1}{2}, \quad f'(2) = -\frac{1}{2^2}, \quad f''(2) = 2!\frac{1}{2^3}, \quad f'''(2) = -3!\frac{1}{2^4}, \ldots f^{(n)}(2) = (-1)^n n! \frac{1}{2^{n+1}}, \ldots.$$

Hence the Taylor series is

$$f(2) + f'(2)(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \cdots + \frac{f^{(n)}(2)}{n!}(x-2)^n + \cdots$$

$$= \frac{1}{2} - \frac{(x-2)}{2^2} + \frac{(x-2)^2}{2^3} - \cdots + (-1)^n \frac{(x-2)^n}{2^{n+1}} + \cdots .$$

This is a geometric series with first term $1/2$ and common ratio $r = -(x-2)/2$. It converges absolutely for $|x-2| < 2$. The sum is

$$\frac{1/2}{1 + (x-2)/2} = \frac{1}{x}.$$

Thus the Taylor series generated by $f(x) = 1/x$ at $a = 2$ converges to $1/x$ for $|x-2| < 2$ or $0 < x < 4$.

Example

The function

$$f(x) = \begin{cases} 0, & x = 0 \\ e^{-1/x^2}, & x \neq 0 \end{cases}$$

has derivatives of all orders at $x = 0$. Indeed, $f^{(n)}(0) = 0$ for all n.

Thus the Taylor series generated by f at $x = 0$ is

$$f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots$$

$$= 0 + 0 \cdot x + 0 \cdot x^2 + \cdots 0 \cdot x^n + \cdots$$

$$= 0 + 0 + \cdots + 0 + \cdots .$$

The series converges for every x (the sum is 0) but converges to $f(x)$ only at $x = 0$.

Homework

Find the Taylor series generated by the following functions at $x = 0$:

1. $f(x) = e^x$
2. $f(x) = \sin x$
3. $f(x) = \cos x$

Also prove that, in each case, the Taylor series converges to the corresponding function for all x.